1. SINGLE INHERITANCE

Objective : To write a Java program to demonstrate **Single Inheritance**.

Algorithm :

- Start the program.
- Create a superclass A with a method sum(int x) that accepts an integer and displays its value.
- Create a subclass B using extends A, so that B inherits the sum() method from A.
- Define a method sub(int y) inside class B to display the value of y.
- In the main() method, create an object obj of class B.
- Use obj.sum(10) to call the inherited method from class A.
- Use obj.sub(20) to call the method defined in class B.
- Stop the program.

UML DIAGRAM :

```
┌──────────────────────┐
│      A               │
├──────────────────────┤
│                      │
├──────────────────────┤
│ + sum(int x) : void  │
└──────────┬───────────┘
           │
           │ extends
           ▼
┌──────────────────────┐
│          B           │
├──────────────────────┤
│                      │
├──────────────────────┤
│ + sub(int y) : void  │
│ + main(String[])     │
```

SOURCE CODE :

```
class A{
   void sum(int x){
      System.out.println("the value of x is :"+x);
   }
}
public class B extends A{

   void sub(int y){
      System.out.println("the value of y is :"+(y));
   }
   public static void main(String[] args) {
      B obj = new B();
      obj.sum(10);
      obj.sub(20);
   }
}
```

OUTPUT :

```
the value of x is :10
the value of y is :20
```

2. Hierarchical Inheritance

Objective

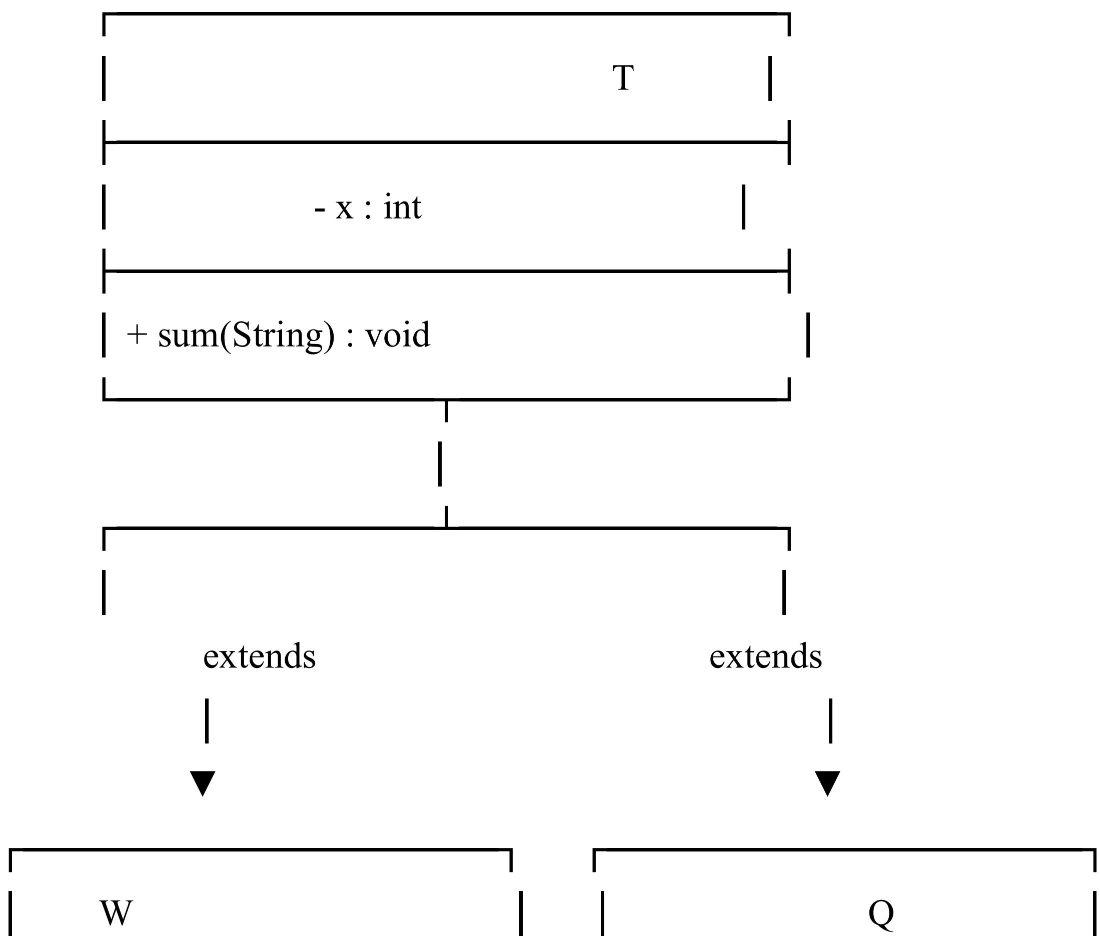
To write a Java program to demonstrate **Hierarchical Inheritance**.

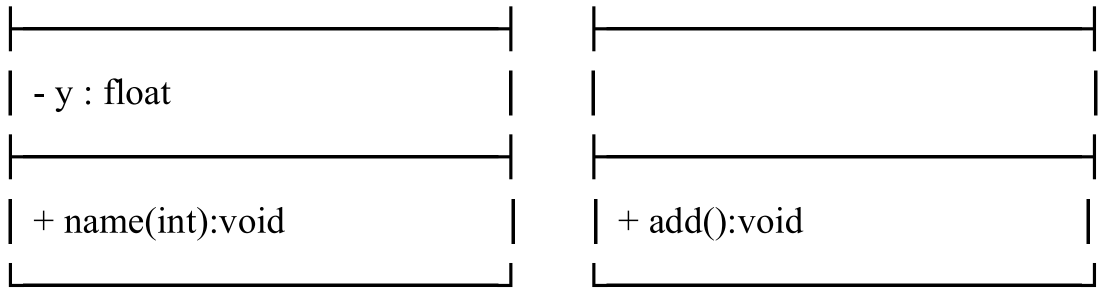
Algorithm

- Start the program.
- Create a superclass T with a variable x and a method sum(String name) to display the name of a person.

- Create a subclass W using extends T, so that W inherits the properties and methods of class T.
- Define a variable y and a method name(int r) inside class W to display the value of r.
- Create another subclass Q using extends T, so that Q also inherits the properties and methods of class T.
- Define a method add() inside class Q to display a message about hierarchical inheritance.
- In the main() method, create an object q of class Q and an object w of class W.
- Use q.add() to call the method defined in class Q.
- Use w.name(34) to call the method defined in class W.
- Use q.sum("manodhara") to call the inherited method from class T.
- Stop the program.

UML DIAGRAM :





SOURCE CODE:

```
class T{
   int x;
   void sum(String name){
      System.out.println("the name of person is :"+name);
   }
}
class W extends T{
   float y=23.45f;
   void name(int r){
      System.out.println("the value of r is:"+r);
   }
}
class Q extends T{
   void add(){
      System.out.println("this is heirarichal inheritance");
   }
}

public class Heirarichal {
   public static void main(String[] args) {
      Q q=new Q();
      W w=new W();
      q.add();
      w.name(34);
      q.sum("manodhara");
   }
}
```

OUTPUT :

```
this is heirarichal inheritance
the value of r is:34
the name of person is :manodhara
```

3.MULTILEVEL INHERITANCE

Objective

To write a Java program to demonstrate **Multilevel Inheritance**.

Algorithm

- Start the program.
- Create a superclass M with a variable x initialized to 98.43 and a method mul(int y) to calculate and display the multiplication of x and y.
- Create a subclass Y using extends M, so that Y inherits the properties and methods of class M.
- Define a method name(String name) inside class Y to display the student's name and percentage.
- Create another subclass R using extends Y, so that R inherits the properties and methods of class Y and indirectly from class M.
- Define a method div() inside class R to display a message about multilevel inheritance.
- In the main() method, create an object man of class R.
- Use man.mul(2) to call the inherited method from class M.
- Use man.name("manodhara") to call the method inherited from class Y.
- Use man.div() to call the method defined in class R.
- Stop the program.

UML DIAGRAM :

M

- x : float = 98.43

+ mul(int y) : void

extends

Y

+ name(String) : void

extends

R

+ div() : void

SOURCE CODE :

```
class M{
   float x=98.43f;
   void mul(int y) {
      System.out.println("the mul is :" + (x * y));
   }
}
class Y extends M{

   void name(String name){
      System.out.println("the student "+name+ " secured "+x+" percentage");
      System.out.println(x);
   }
}
class R extends Y{
   void div(){
      System.out.println("this is multilevel inheritance");

   }
}
public class MultiLevel {
  public  static void main(String[] args) {
      R man = new R();
     man.mul(2);
     man.name("manodhara");
     man.div();

   }
}
```

OUTPUT :

```
the mul is :196.86
the student manodhara secured 98.43 percentage
98.43
this is multilevel inheritance
```

4.HYBRID INHERITANCE :

Objective

To write a Java program to demonstrate **Hybrid Inheritance** .

ALGORITHM :

- Start the program.
- Import the Scanner class to read input from the user.
- Create a superclass Bb with a Scanner object, an integer variable a, and a method add() to read and display the value of a.
- Create a subclass Ss using extends Bb, so that Ss inherits the properties and methods of class Bb.
- Define an integer variable b and a method sum() inside class Ss to read the value of b and calculate the sum of a and b.
- Create a subclass Ww using extends Ss, so that Ww inherits the properties and methods of class Ss and Bb.
- Define a method eds() inside class Ww to display a message about hybrid inheritance.
- Create another subclass Rr using extends Ss, so that Rr also inherits the properties and methods of class Ss and Bb.
- Define a method sub() inside class Rr to calculate and display the difference between a and b.
- In the main() method, create an object re of class Ww and an object ew of class Rr.
- Use re.add() to read and display the value of a through the inherited method from class Bb.
- Use re.sum() to read and display the value of b and calculate the sum of a and b.
- Use re.eds() to call the method defined in class Ww.
- Use ew.add() to call the inherited add() method from class Bb.
- Use ew.sum() to call the inherited sum() method from class Ss.
- Use ew.sub() to calculate and display the difference between a and b.
- Stop the program.

UML DIAGRAM :

```
┌──────────────────────┐
│        Bb        │
├────────────────────────┤
│ - sc : Scanner       │
│ - a : int        │
├────────────────────────┤
│ + add() : void       │
└───────────┬───────────┘
     │
    extends
     │
     ▼
┌──────────────────────┐
│        Ss                  │
├────────────────────────┤
│ - b : int                   │
├────────────────────────┤
│ + sum() : void                │
└───────────┬───────────┘
            │
┌───────────┴───────────┐
  │          │
 extends        extends
   │               │
   ▼               ▼
┌────────────────┐  ┌────────────────┐
```

```
|    Ww                |   |    Rr              |
├──────────────────────┤   ├──────────────────────┤
|                      |  |                       |
├──────────────────────┤   ├──────────────────────┤
| + eds() : void       |  | + sub() : void       |
└──────────────────────┘  └──────────────────────┘
```

SOURCE CODE :

```
import java.util.Scanner;


class Bb{
   Scanner sc=new Scanner(System.in);
   int a;
   void add(){
      System.out.println("enter the value of a :");
       a=sc.nextInt();
      System.out.println("the value of a is :"+a);
      System.out.println("this is a method in super class ");
   }
}
class Ss extends Bb{
  int b;
   void sum(){
      System.out.println("enter the value of b :");
       b=sc.nextInt();
      System.out.println("the value of b is :"+b);
      System.out.println("the sum of a and b is :"+(a+b));
   }
}
```

```
class Ww extends Ss{
    void eds(){
        System.out.println("this is hybrid inheritance");
    }
}
class Rr extends Ss{
    void sub(){
        System.out.println("the diff between a and b is :"+(a-b));
    }
}
public class Hybrid {
    public static void main(String[] args) {
Ww re=new Ww();
Rr ew=new Rr();
re.add();
re.sum();
re.eds();
ew.add();
ew.sum();
ew.sub();
    }
}
```

OUTPUT :

```
enter the value of a :
3
the value of a is :3
this is a method in super class
enter the value of b :
5
the value of b is :5
the sum of a and b is :8
this is hybrid inheritance
```